

DRAWING LIST FOR TEST PACKAGE

PAC-509

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	509	4FA	OD90-PAC-HP2-N-6513-01	1 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
2	509	4FA	OD50-PAC-HP2-N-6509-01	1 OF 1	1	CJ-M15W14-PR-A20-00151	1A	
3	509	4FA	OD32-PAC-HP2-N-6524-01	1 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
4	509	4FA	OD32-PAC-HP2-N-6524-02	2 OF 2	1	CJ-M15W14-PR-A20-00151	1A	
5	509	4FA	OD50-PAC-HP2-N-6510-01	1 OF 1	1	CJ-M15W14-PR-A20-00151	1A	
6	509	4FA	OD50-PAC-HP2-N-6511-01	1 OF 1	0	CJ-M15W14-PR-A20-00151	1A	
7	509	4FA	OD90-PAC-HP2-N-6513-02	2 OF 2	0	CJ-M15W14-PR-A20-00151	1A	
8	509	5FA	OD90-PAC-HP2-N-6513-01	1 OF 1	0	CJ-M15W14-PR-A20-00151	1A	
9	509	RFA	OD90-PAC-HP2-E-6513-01	1 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
10	509	RFA	OD90-PAC-HP2-E-6513-02	2 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
11	509	RFA	OD32-PAC-HP2-E-6517-01	1 OF 2	1	CJ-M15W14-PR-A20-000105/151	1A	
12	509	RFA	OD32-PAC-HP2-E-6517-02	2 OF 2	1	CJ-M15W14-PR-A20-000105/151	1A	
13	509	RFA	OD32-PAC-HP2-E-6558-01	1 OF 1	1	CJ-M15W14-PR-A20-000105	1A	
14	509	RFA	OD90-PAC-HP2-E-6513-03	3 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
15	509	RFA	OD32-PAC-HP2-E-6516-01	1 OF 2	1	CJ-M15W14-PR-A20-000104/151	1A	
16	509	RFA	OD32-PAC-HP2-E-6516-02	2 OF 2	1	CJ-M15W14-PR-A20-000104151	1A	
17	509	RFA	OD32-PAC-HP2-E-6556-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
18	509	RFA	OD32-PAC-HP2-E-6514-01	1 OF 2	1	CJ-M15W14-PR-A20-000102/151	1A	
19	509	RFA	OD32-PAC-HP2-E-6514-02	2 OF 2	1	CJ-M15W14-PR-A20-000102/151	1A	
20	509	RFA	OD32-PAC-HP2-E-6552-01	1 OF 1	1	CJ-M15W14-PR-A20-000102	1A	

DRAWING LIST FOR TEST PACKAGE

PAC-509

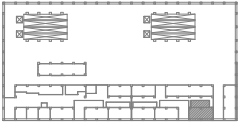
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	509	RFA	OD32-PAC-HP2-E-6515-01	1 OF 2	1	CJ-M15W14-PR-A20-000151	1A	
22	509	RFA	OD32-PAC-HP2-E-6515-02	2 OF 2	1	CJ-M15W14-PR-A20-000151	1A	
23	509	RFA	OD32-PAC-HP2-E-6554-01	1 OF 1	1	CJ-M15W14-PR-A20-000103	1A	
24	509	RFA	OD32-PAC-HP2-E-6523-01	1 OF 2	1	CJ-M15W14-PR-A20-000110/151	1A	
25	509	RFA	OD32-PAC-HP2-E-6523-02	2 OF 2	1	CJ-M15W14-PR-A20-000110/151	1A	
26	509	RFA	OD32-PAC-HP2-E-6566-01	1 OF 1	1	CJ-M15W14-PR-A20-000110	1A	
27	509	RFA	OD90-PAC-HP2-E-6513-04	4 OF 4	1	CJ-M15W14-PR-A20-00151	1A	
28	509	RFA	OD32-PAC-HP2-E-6522-01	1 OF 2	1	CJ-M15W14-PR-A20-000109/151	1A	
29	509	RFA	OD32-PAC-HP2-E-6522-02	2 OF 2	1	CJ-M15W14-PR-A20-000109/151	1A	
30	509	RFA	OD32-PAC-HP2-E-6564-01	1 OF 1	1	CJ-M15W14-PR-A20-000109	1A	
31	509	RFA	OD32-PAC-HP2-E-6520-01	1 OF 2	1	CJ-M15W14-PR-A20-000107/151	1A	
34	509	RFA	OD32-PAC-HP2-E-6521-01	1 OF 2	1	CJ-M15W14-PR-A20-000108/151	1A	

ITEM NO.	W14-T-PAC-4605 A		
SERVICE	PAC STORAGE TANK		
Q'TY	1		
TYPE	VERTICAL & CYLINDRICAL		
SIZE	ID x H	m	4.0 x 4.5 (EH/4.2) 1A
CAPA.	m3		57 (EH/53)
MAT'L	FRP		

대외비

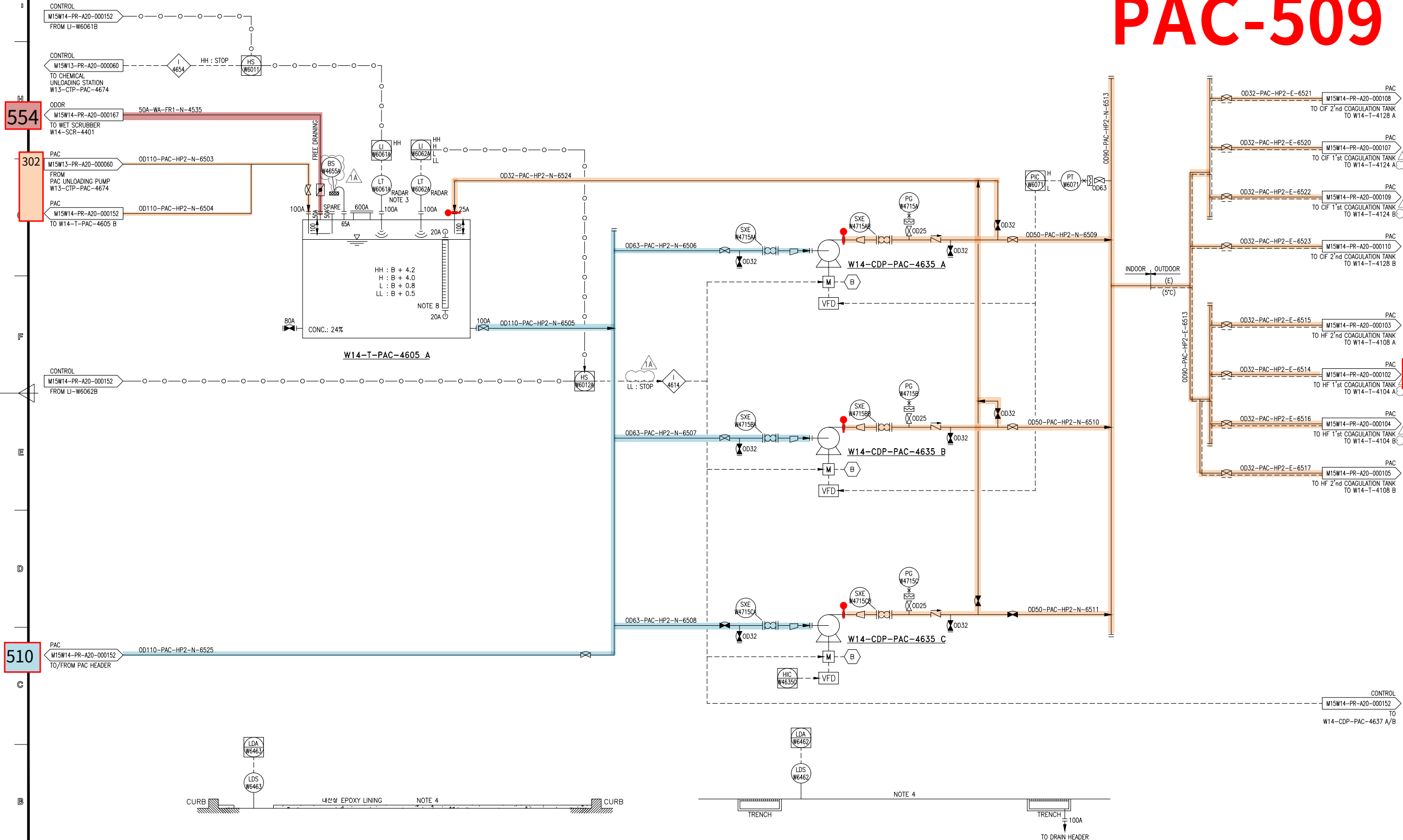


KEY PLAN (4F)

NOTE

- CENTRIFUGAL PUMP
MAT'L 표기 : CASING/IMPELLER
- 약품종류에 따라 LINE 색상을 구분하여야 한다.
- CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
- CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 되는 DRAIN은 인근의 TRENCH로 접속되어, MAIN HEADER로 이동한다.
- INTERLOCK NO. I-4654
(W13-CTP-PAC-4674 AUTO STOP)
REMOTE STATUS에서 W14-T-PAC-4605 A/B
VEL (U-W6061 A/B) HIGH HIGH STOP.
- INTERLOCK NO. I-4614
(W14-CDP-PAC-4635 A/B/C
W14-CDP-PAC-4637 A/B AUTO STOP)
REMOTE STATUS에서 W14-T-PAC-4605 A/B
VEL (U-W6062 A/B) LOW LOW STOP.
시 CLOSE 되는 배관에는 LOTO 설치.
TANK의 역변 레벨 개시지는 VENDOR에서 공급한다.

PAC-509



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1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE

제도	이인학
DRAWN BY	
검토	한진희
CHECKED BY	
승인	박동명
APPROVED BY	
제작사	

SK 에코플랜트

공사명	PROJECT TITLE
M15X Ph4 WWT Project	

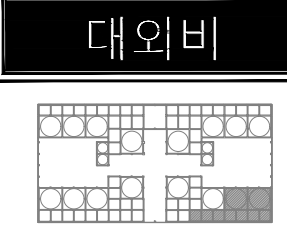
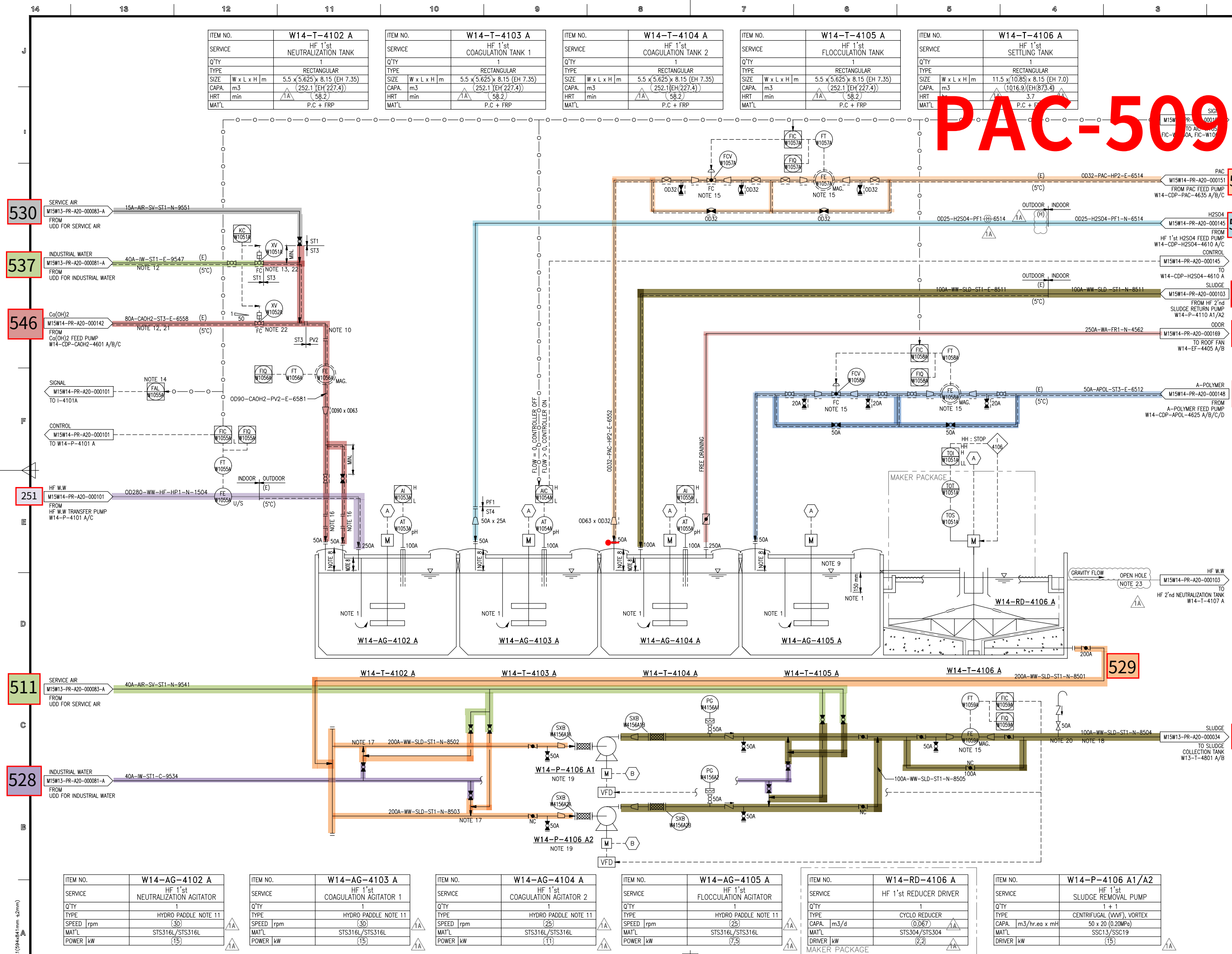
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM PAC STORAGE TANK & FEED PUMP (1/2)	

축척	SCALE
NONE	
날짜	DATE
2025. 08. 14	

PROJECT NO.	22330D	REV.	
도면번호	CJ-M15W14-PR-A20-000151		
SHEET NO.			

1A

ITEM NO.	W14-CDP-PAC-4635 A/B/C
SERVICE	PAC FEED PUMP
Q'TY	2 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr.ea x MPa
4 x 0.46	
MAT'L	FRP/FRP
POWER kW	11 1A



KEY PLAN (5F)

- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치할 것.
 2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
 3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
사전 시 CLOSE 되는 배관에는 LOTO 설치.
 4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 할 것.
AB 으로부터 100mm 떨어진 것
 7. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
SS304 + FRP LINING 적용.
 8. 11. VENDOR에서 결정한다.
 9. 12. 라인은 상부에서 시수 도입하도록 설치한다.
W1051A OPEN (12'간마다 15') 일 경우 :
XV-W1052A CLOSE
 10. 14. W14-P-4101 A가 RUN 후 30초 후부터 동작한다.
 11. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 12. 16. 구간 투명 PVC 적용. (주입 여부 확인용)
 13. 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 14. 18. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
 15. 19. SLUDGE PUMP는 TIMER에 의해 운전함.
 16. 20. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 17. 21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 위험 최소화
및 주입배관 고제를 용이하게 한다.
 18. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 19. 23. CONCRETE WALL OPENING을 통한 유체 이송
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

수정	REV.	실명	DESCRIPTION	날짜	DATE
제	도	이인학			
검	토	한진희			
인	검	박동명			
제	작	사			

SK 에코플랜트	
공 사 명	
PROJECT TITLE	
M15X Ph4 WWT Project	
도면명	
DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM	
HF 1st COAGULATION & SETTLING	
TANK A (1/2)	
속	척
SCALE	NONE
날	짜
DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000102
SHEET NO.	1A

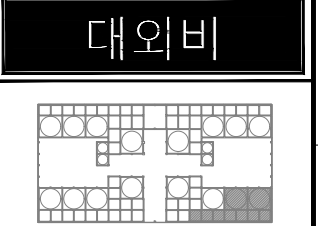
ITEM NO.	W14-T-4107 A
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (252.1 (EH 201.1))
HRT	min (51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 A
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (252.1 (EH 201.1))
HRT	min (51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 A
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.7 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3 (261.3 (EH 208.4))
HRT	min (53.4)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 A
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 11.7 x 10.85 x 8.15 (EH 6.5)
CAPA.	m3 (1034.6 (EH 825.1))
HRT	hr 3.5
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304



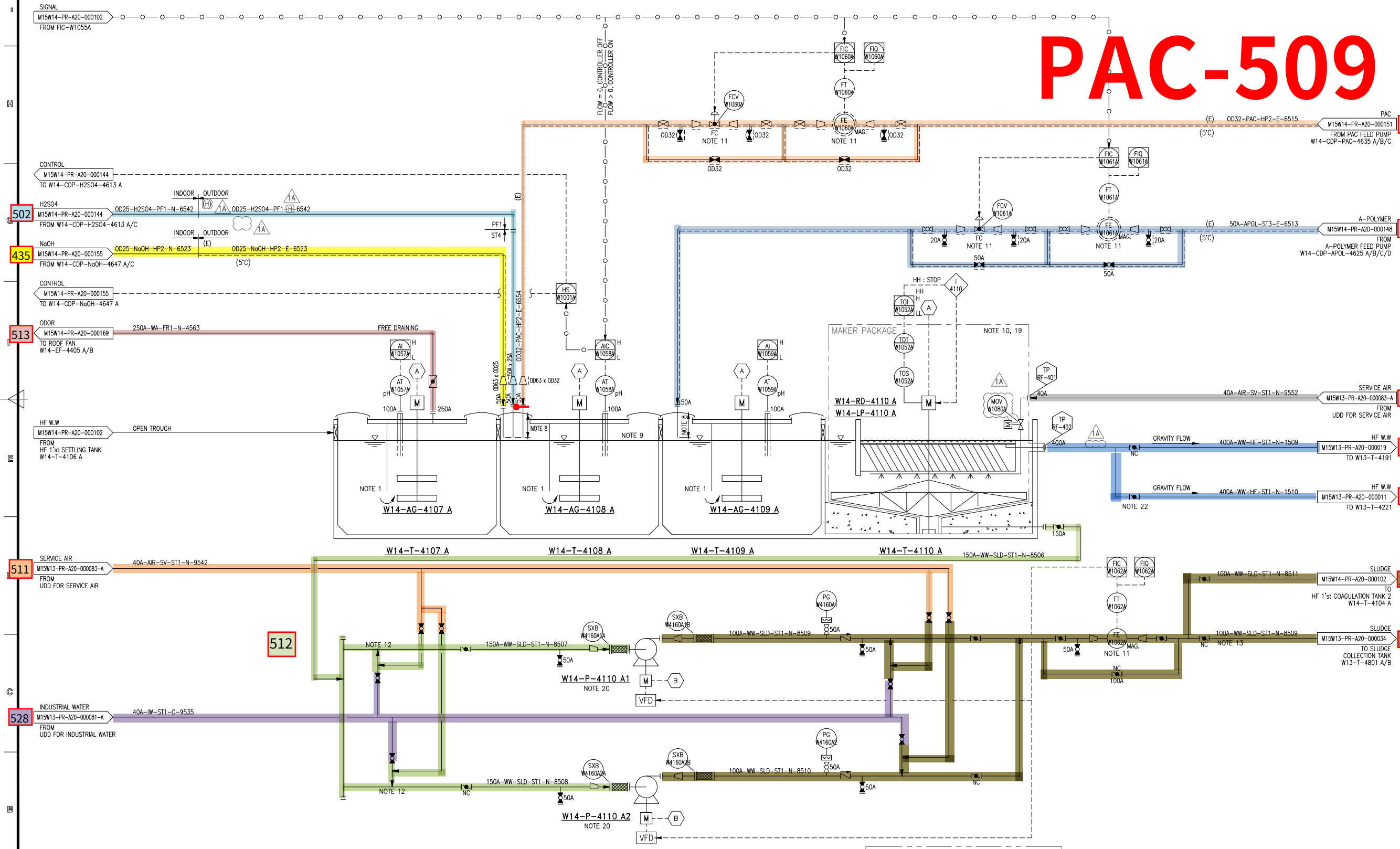
KEY PLAN (5F)

- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온천 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 거주층의 상계사창은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 검사한 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. VALVE 이용이 접근 가능해야 하며, LIMIT SWITCH가 육안에 보이기 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
정	실	날
REV.	DESCRIPTION	DATE
제	도	인
DRAWN BY	이	인
검	토	한
CHECKED BY	진	희
승	인	박
APPROVED BY	동	영
제	작	사

공	사	명
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
HF 2'nd COAGULATION & SETTLING		
TANK A (1/2)		
속		
SCALE		
NONE		
날		
DATE		
2025. 08. 14		
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000103	1A

PAC-509



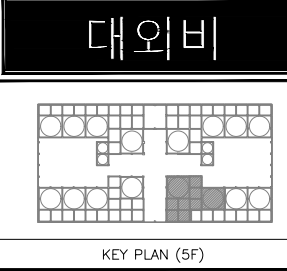
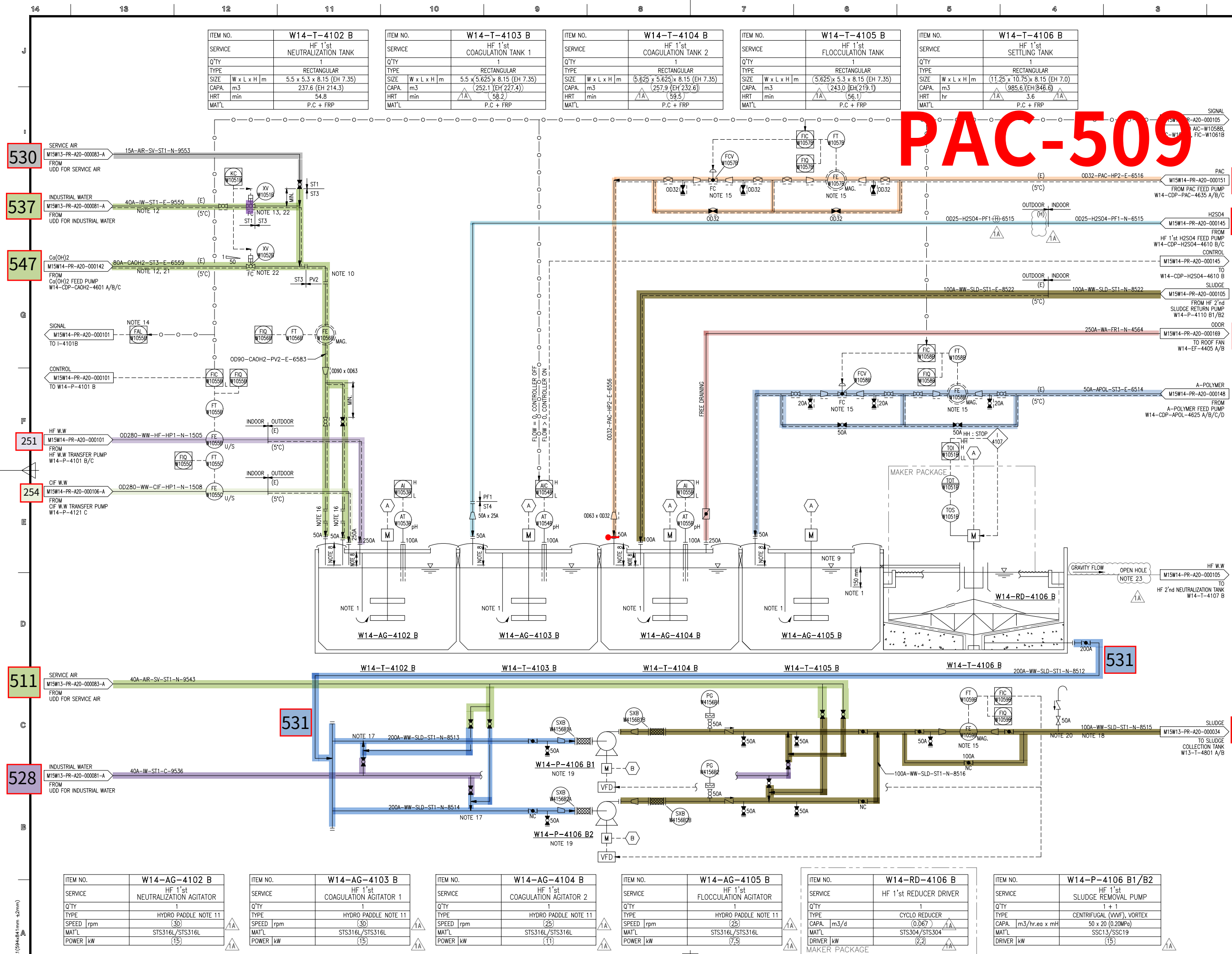
ITEM NO.	W14-AG-4107 A
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4108 A
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (11)

ITEM NO.	W14-AG-4109 A
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-RD-4110 A
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4110 A1/A2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)



- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 1개 이상.
 3. 2.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. INHOLE : 2개 이상 설치한다.
INHOLE COVER : FRP
시 CLOSE 되는 배관에는 LOTO 설치.
 5. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 6. AGITATOR MAT'L 표기 : SHAFT/BLADE
 7. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
B 으로부터 100mm 떨어질 것
 8. FRP LINING TANK의 SLEEVE는 504 + FRP LINING 적용.
 9. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
 11. 주용 VENDOR에서 결정한다.
 12. 라인은 상부에서 시공 시 유입하드록 설치한다.
 13. W1051B OPEN (12시간마다 15초) 일 경우 : W1052B CLOSE
 14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 16. 구간 투명 PVC 적용. (주입 여부 확인용)
 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.



1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인
DRAWN BY	한 진 희	
제	작	사
SK 에코플랜트		
공	사	명
PROJECT TITLE		
M15X Ph4 WWT Project		
도	명	
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
HF 1st COAGULATION & SETTLING		
TANK B (2/2)		
속	척	
SCALE	NONE	
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000104	1A
SHEET NO.		

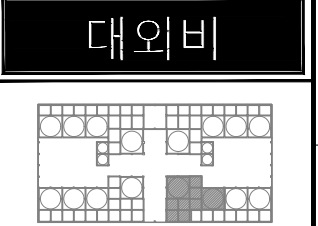
ITEM NO.	W14-T-4107 B	
SERVICE	HF 2'nd NEUTRALIZATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	(5.75 x 5.45) x 8.15 (EH 6.5)
CAPA.	m3	(255.4 (EH 203.7)
HRT	min	(52.1)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4108 B	
SERVICE	HF 2'nd COAGULATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	(5.75 x 5.175) x 8.15 (EH 6.5)
CAPA.	m3	(242.5 (EH 193.4)
HRT	min	(49.5)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4109 B	
SERVICE	HF 2'nd FLOCCULATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	(5.625 x 5.175) x 8.15 (EH 6.5)
CAPA.	m3	(237.2 (EH 189.2)
HRT	min	(48.4)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4110 B	
SERVICE	HF 2'nd SETTLING TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	(11.375 x 10.85) x 8.15 (EH 6.5)
CAPA.	m3	(1005.9 (EH 802.2)
HRT	hr	3.4
MAT'L	P.C + FRP	

ITEM NO.	W14-LP-4110 B	
SERVICE	LAMELLAR PLATE	
Q'TY	1	
CAPA.	W x L x H m	BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)	
MAT'L	ABS+SS304	



KEY PLAN (5F)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
HOLE COVER : FRP
 5. 시 CLOSE 되는 배관에는 LOTO 설치.
 6. TRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 특수용의 설계사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 TROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 청소 및 유지관리시설을 PACKAGE VENDOR 공급받아야 함.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. 용이 접근 가능해야 하며, FITCH가 육안에 보이게 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
375	APPROVAL FOR CONSTRUCTION	24.09.30
375	실명 DESCRIPTION	날짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	
제 작 사	

SK 에코플랜트

공 사 명
PROJECT TITLE

M15X Ph4 WWT Project

도면명
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM
HF 2'nd COAGULATION & SETTLING
TANK B (2/2)

속 척
SCALE

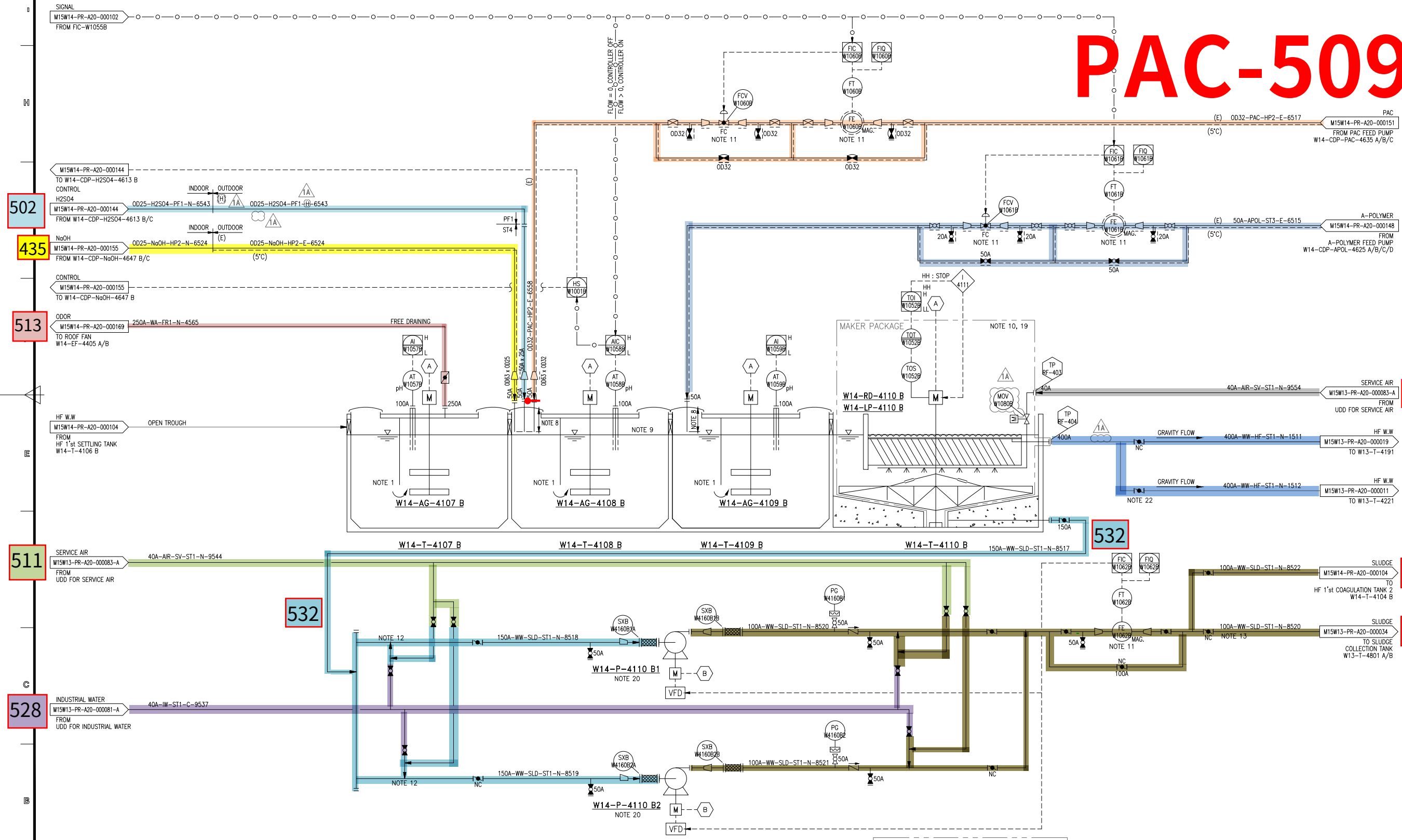
NONE

날 짜
DATE

2025. 08. 14

PROJECT NO.	22330D	REV.	
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000105	1A	

PAC-509



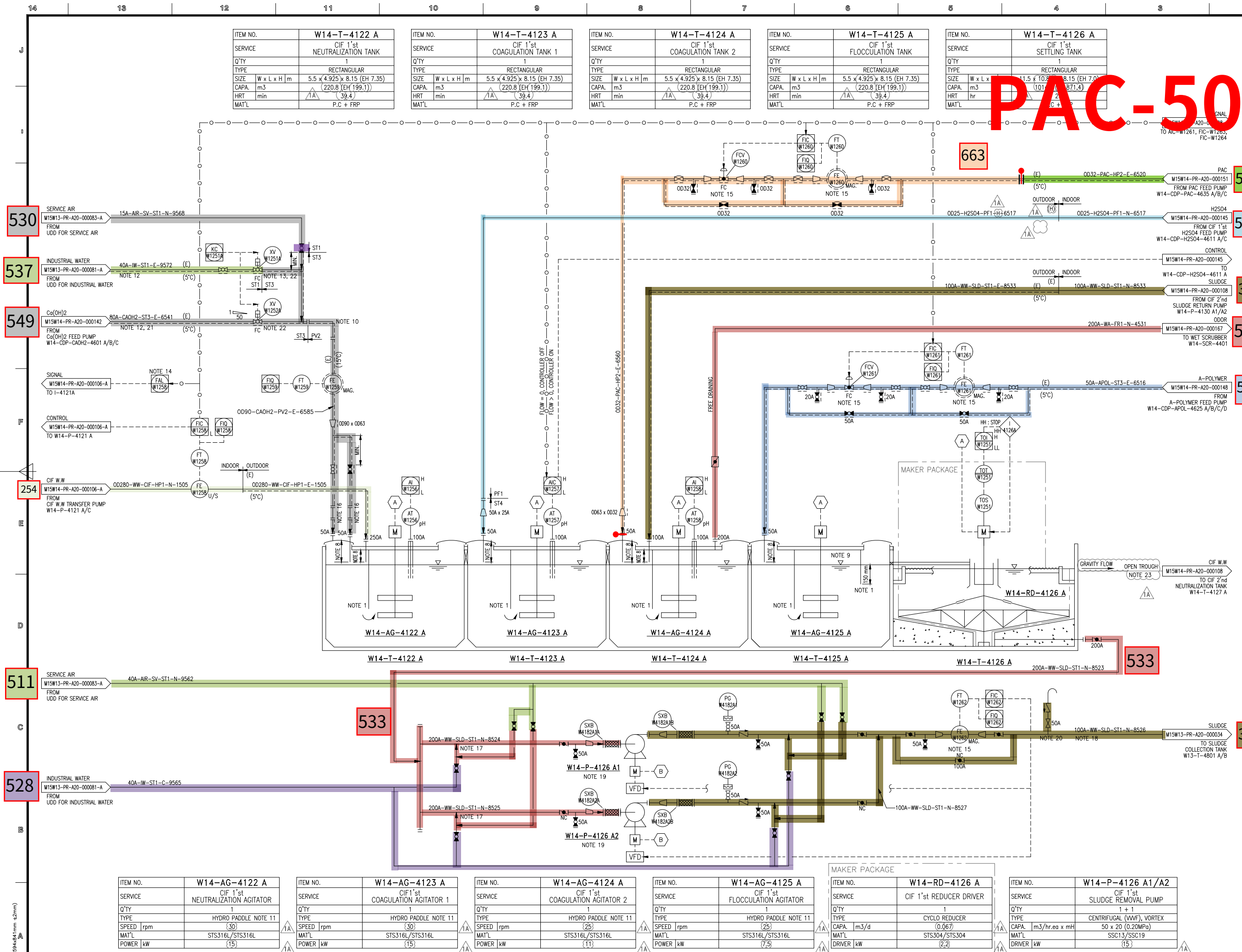
ITEM NO.	W14-AG-4107 B	
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR	
Q'TY	1	
TYPE	HYDRO PADDLE NOTE 15	
SPEED	rpm	(30)
MAT'L	STS316L/STS316L	
POWER	kW	
	(15)	

ITEM NO.	W14-AG-4108 B	
SERVICE	HF 2'nd COAGULATION AGITATOR	
Q'TY	1	
TYPE	HYDRO PADDLE NOTE 15	
SPEED	rpm	(25)
MAT'L	STS316L/STS316L	
POWER	kW	
	(15)	

ITEM NO.	W14-AG-4109 B	
SERVICE	HF 2'nd FLOCCULATION AGITATOR	
Q'TY	1	
TYPE	HYDRO PADDLE NOTE 15	
SPEED	rpm	(25)
MAT'L	STS316L/STS316L	
POWER	kW	
	(7.5)	

ITEM NO.	W14-RD-4110 B	
SERVICE	HF 2'nd REDUCER DRIVER	
Q'TY	1	
TYPE	CYCLO REDUCER	
CAPA.	m3/d	(0.067)
MAT'L	STS304/STS304	
POWER	kW	
	(2.2)	

ITEM NO.	W14-P-4110 B1/B2	
SERVICE	HF 2'nd SLUDGE RETURN PUMP	
Q'TY	1 + 1	
TYPE	CENTRIFUGAL (VVF), VORTEX	
CAPA.	m3/hr.ea x mH	50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19	
POWER	kW	
	(30)	



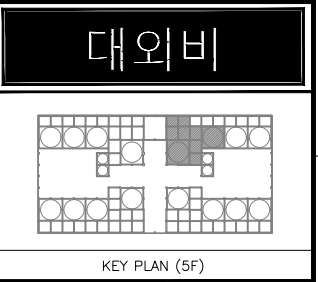
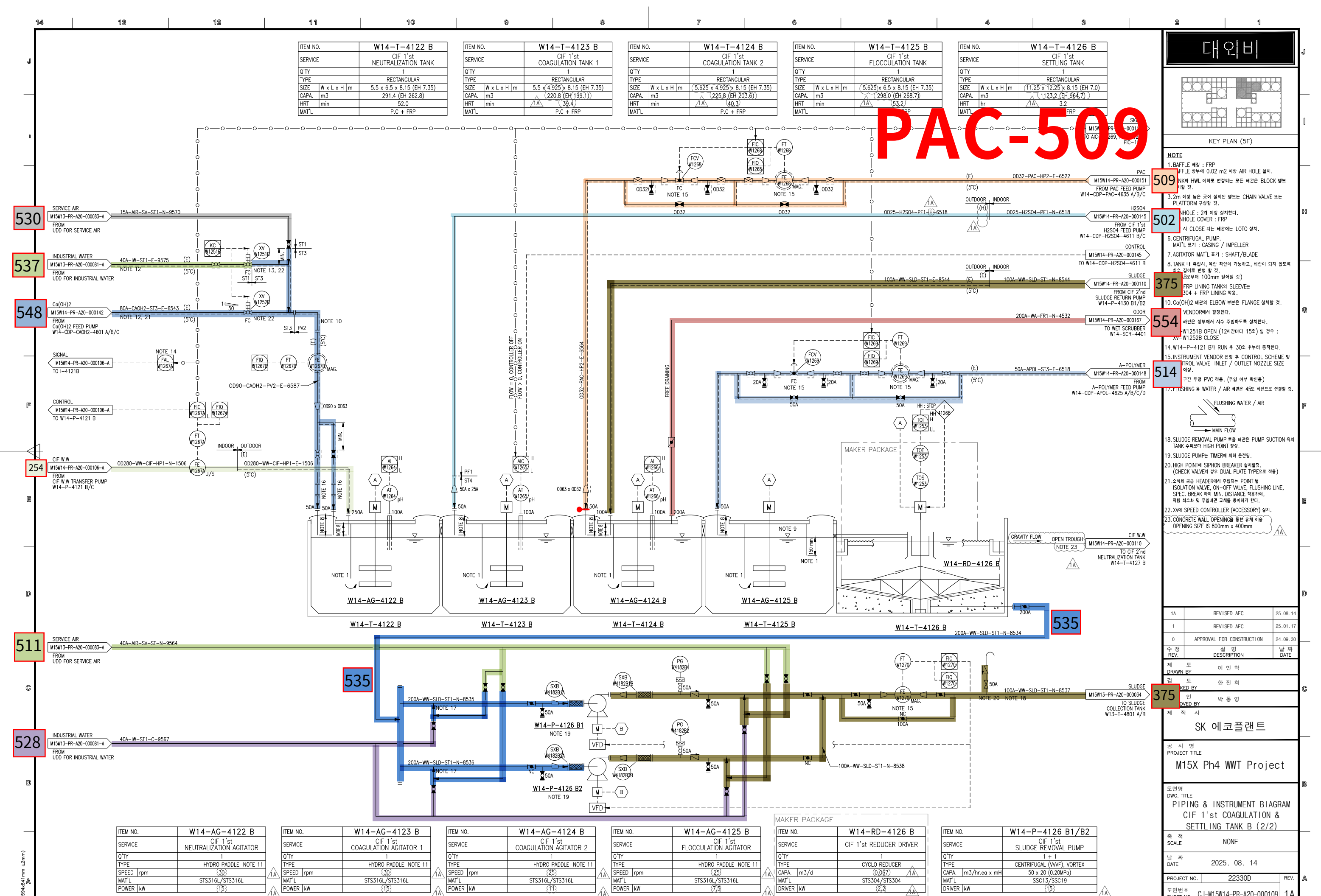
대외비

KEY PLAN (5F)

NOTE

1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
2. NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치.
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
4. HOLE : 2개 이상 설치한다.
HOLE COVER : FRP
5. HOLE CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이를 반영 할 것.
B 으로부터 100mm 떨어질 것
9. FRP LINING TANK의 SLEEVE는
504 + FRP LINING 적용.
10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
VENDOR에서 결정한다.
11. 라인은 상부에서 시수 주입하도록 설치한다.
12. W14-P-4121 A가 RUN 후 30초 후부터 동작한다.
13. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
14. 구간 통행 PVC 적용. (주입 여부 확인용)
15. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
16. FLUSHING WATER / AIR
17. MAIN FLOW
18. SLUDGE REMOVAL PUMP를 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
19. SLUDGE PUMP는 TIMER에 의해 운전됨.
20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여,
적합 최소화 및 주입배관 고체를 용이하게 한다.
22. XPM SPEED CONTROLLER (ACCESSORY) 설치.
23. CONCRETE WALL OPENING을 통한 유체 이용
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	인
DRAWN BY	한진희	
검	토	한진희
ED BY	한진희	
375	인	박동명
VED BY	박동명	
제	작	사
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT CIAGRAM		
CIF 1'st COAGULATION &		
SETTLING TANK A (1/2)		
속		
SCALE		
NONE		
날		
DATE		
2025. 08. 14		
PROJECT NO.		
22330D		
REV.		
도면번호		
SHEET NO.		
CJ-M15W14-PR-A20-000107		
1A		



- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
시설 함.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성함.
 - HOLE : 2개 이상 설치한다.
시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 함.
 8. B로부터 100mm 떨어질 것)
 - FRP LINING TANK의 SLEEVE는
504 + FRP LINING 적용.
 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치함.
 - VENDOR에서 결정한다.
 - 라인은 상부에서 시수 주입하도록 설치한다.
 - W1251B OPEN (12시간마다 15초) 일 경우 :
XV-W1252B CLOSE
 14. W14-P-4121 B가 RUN 후 30초 후부터 동작한다.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET / OUTLET NOZZLE SIZE
예정.
 - 구간 투명 PVC 적용. (주입 여부 확인용)
 17. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결함.
 18. SLUDGE REMOVAL PUMP를 배관한 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
 20. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 21. 소액의 공극 HEADER에 주입되는 POINT
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여,
역침 침식 및 주입배관 고체를 용이하게 한다.
 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 23. CONCRETE WALL OPENING을 통한 유체 이용
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
인	박 동 영
APPROVED BY	
제 작 사	

SK 에코플랜트	
공 사 명	
PROJECT TITLE	
M15X Ph4 WWT Project	
도면명	
DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM	
CIF 1'st COAGULATION &	
SETTLING TANK B (2/2)	
속 척	NONE
SCALE	
날 짜	2025. 08. 14
DATE	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000109
SHEET NO.	1A

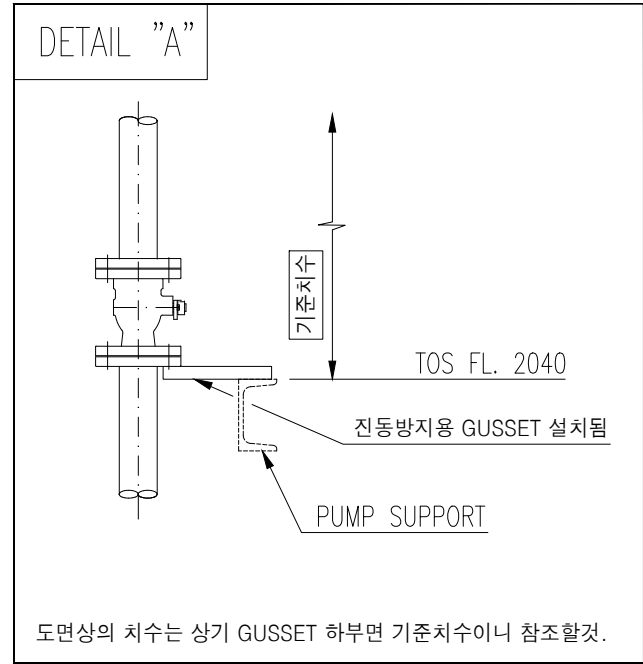
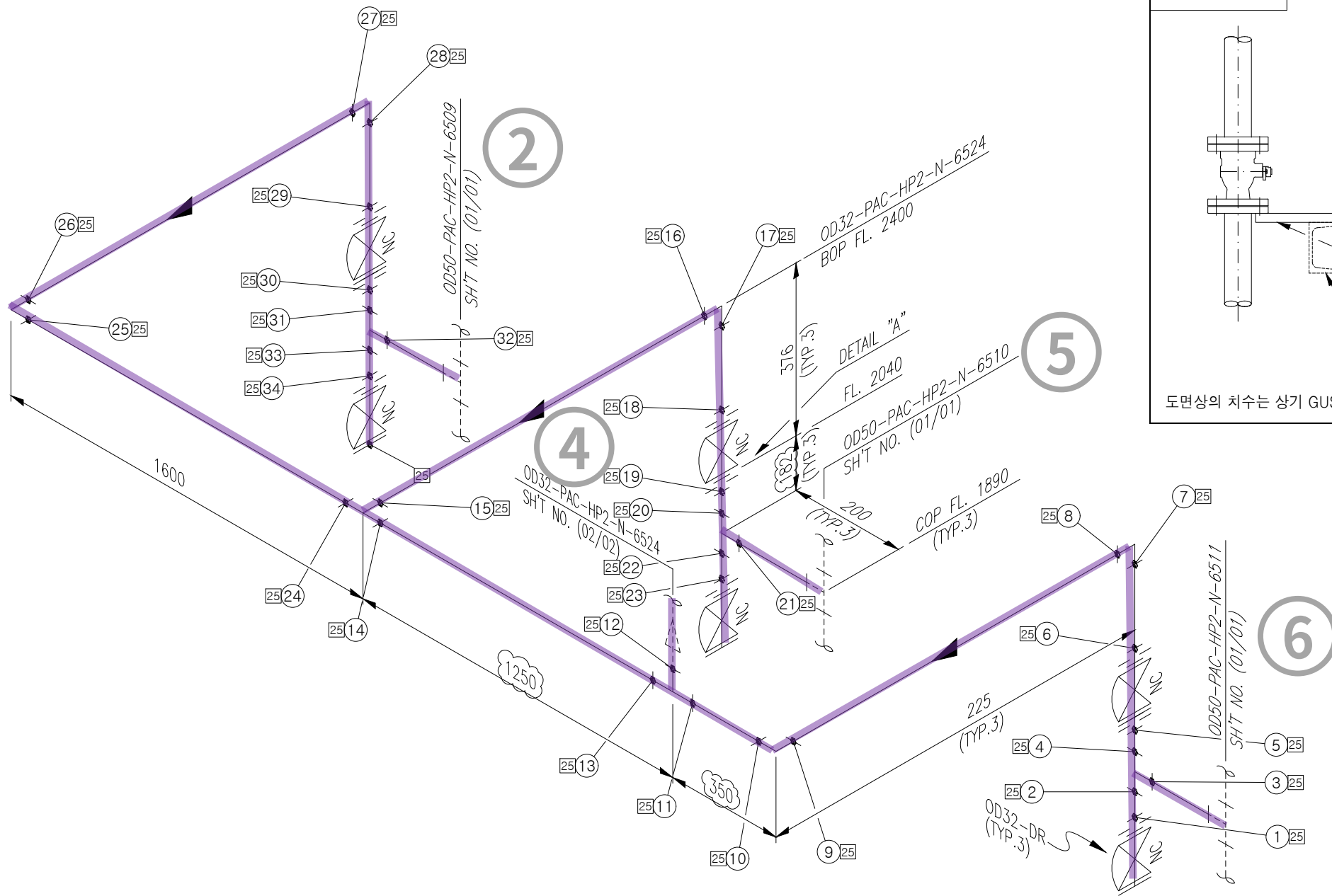
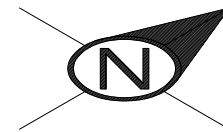


CLIENT



1

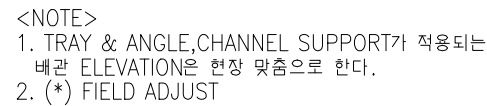
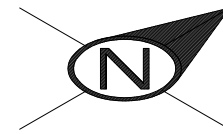
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.4	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD90-PAC-HP2-N-6513	22330D	SH'T NO.	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 02	REV. 1



- <NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
 2. (*) FIELD ADJUST

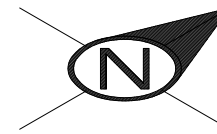
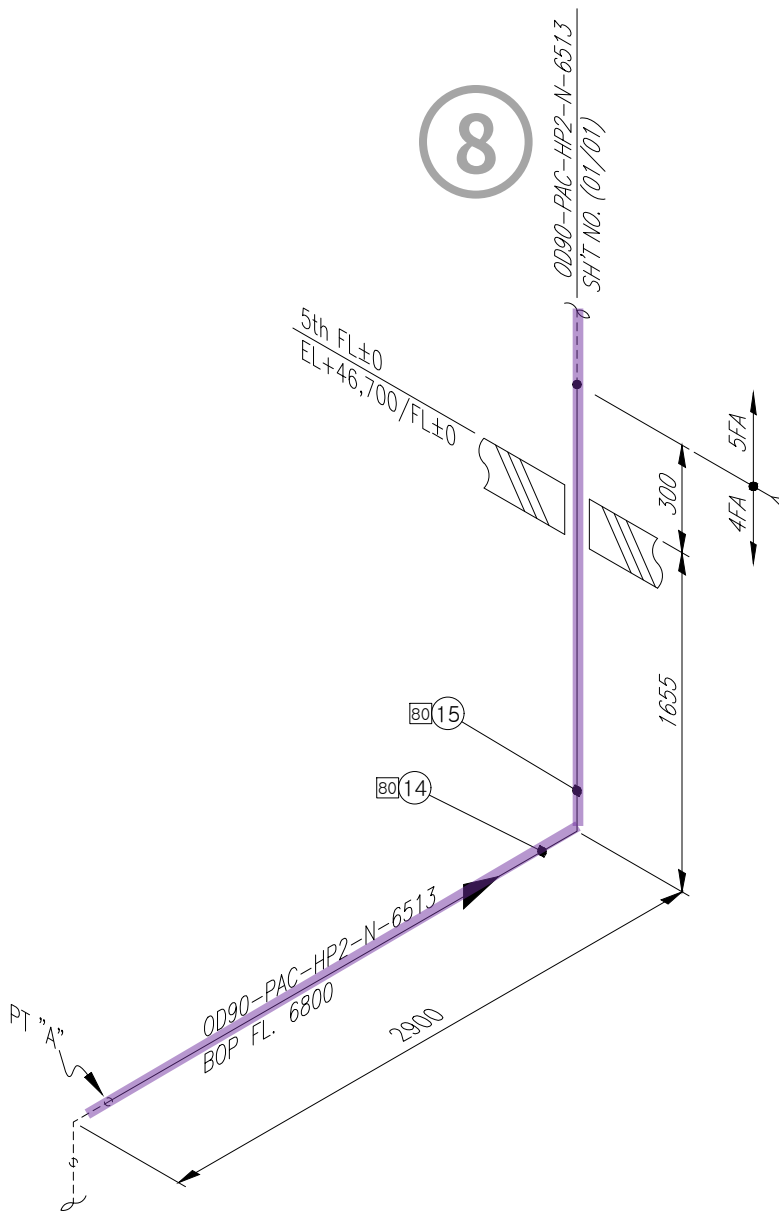
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR 	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-04008-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000151			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD32-PAC-HP2-N-6524	SH'T NO.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 02



CLIENT  

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유포할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



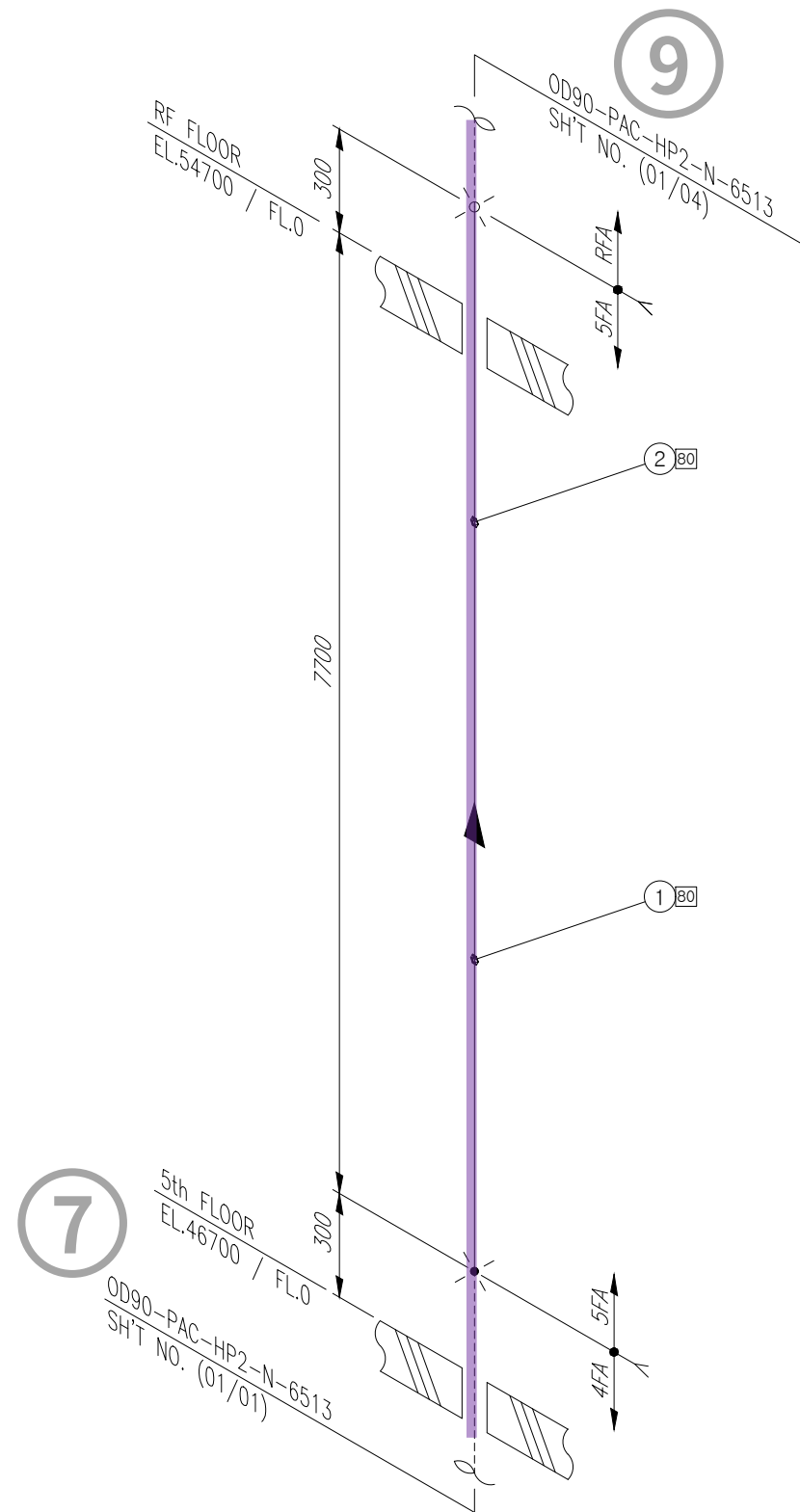
<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

[illegible]

CLIENT		 	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO.	OD90-PAC-HP2-N-6513	SH'T NO. 02 / 02	REV. 0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1												
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	6.3	60	UNIT 4FA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동진
					REF. P & ID CJ-M15W14-PR-A20-000151							
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

[illegible]

<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 		PROJECT NAME	
	THICK.	–	DESIGN	6.3	60	UNIT 5FA									M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY PAC		JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000151	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD90-PAC-HP2-N-6513	SH'T NO. 01 / 01	REV. 0



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W13-PI-A30-R10005-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000151	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'

CLIENT		 	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO. OD90-PAC-HP2-E-6513		SH'T NO. 01 / 04	REV. 1



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CLIENT		 	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY PAC		JOB NO. 22330D	
LINE NO. OD90-PAC-HP2-E-6513		SH'T NO. 02 / 04	REV. 1



1. (*) FIELD ADJUST

CLIENT



11

CONTRACTOR

PROJECT NAME	
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COMMODITY	
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	PAGE

LINE NO. 0D32-PAC-HP2-E-6517

JOB NO.

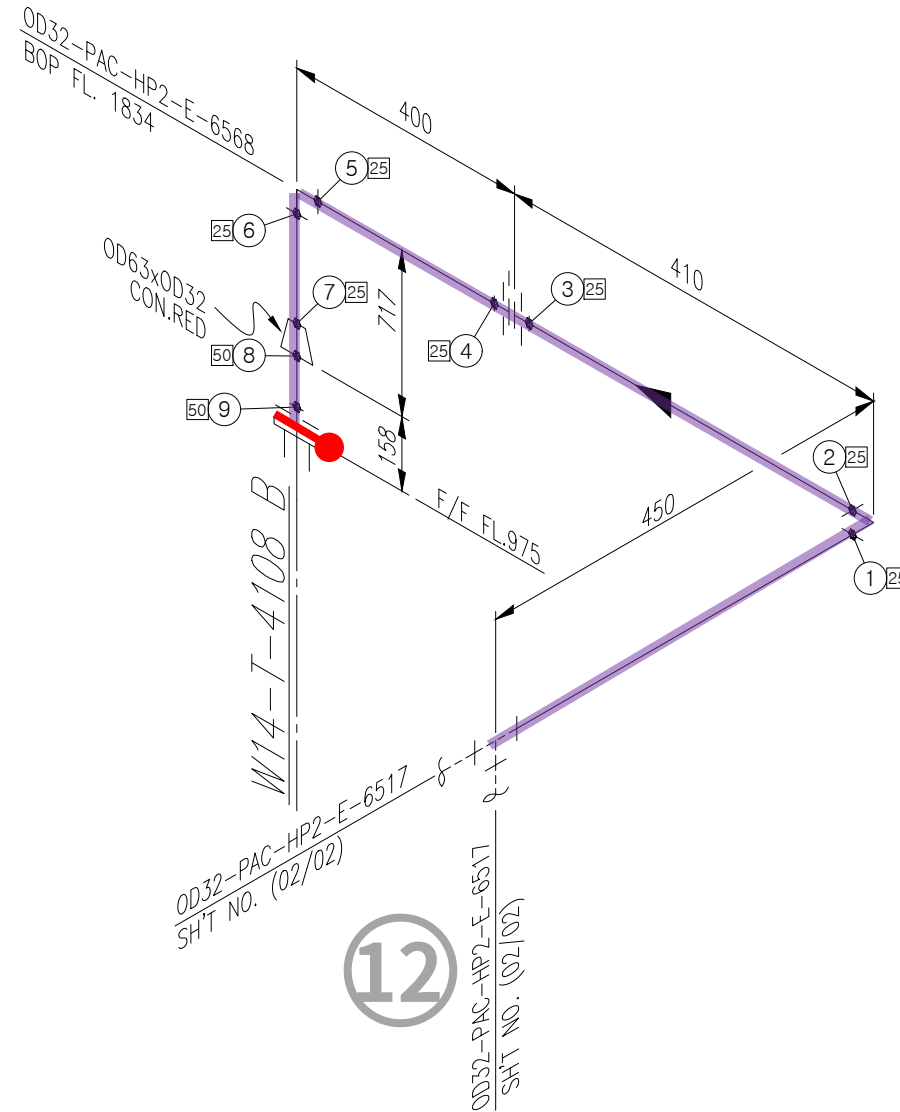
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SH'T NO.	REV.
01 / 03	1



CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	PAC	JOB NO.	
		22330D	
LINE NO.	OD32-PAC-HP2-E-6517	SH'T NO. 02 / 02	REV. 1

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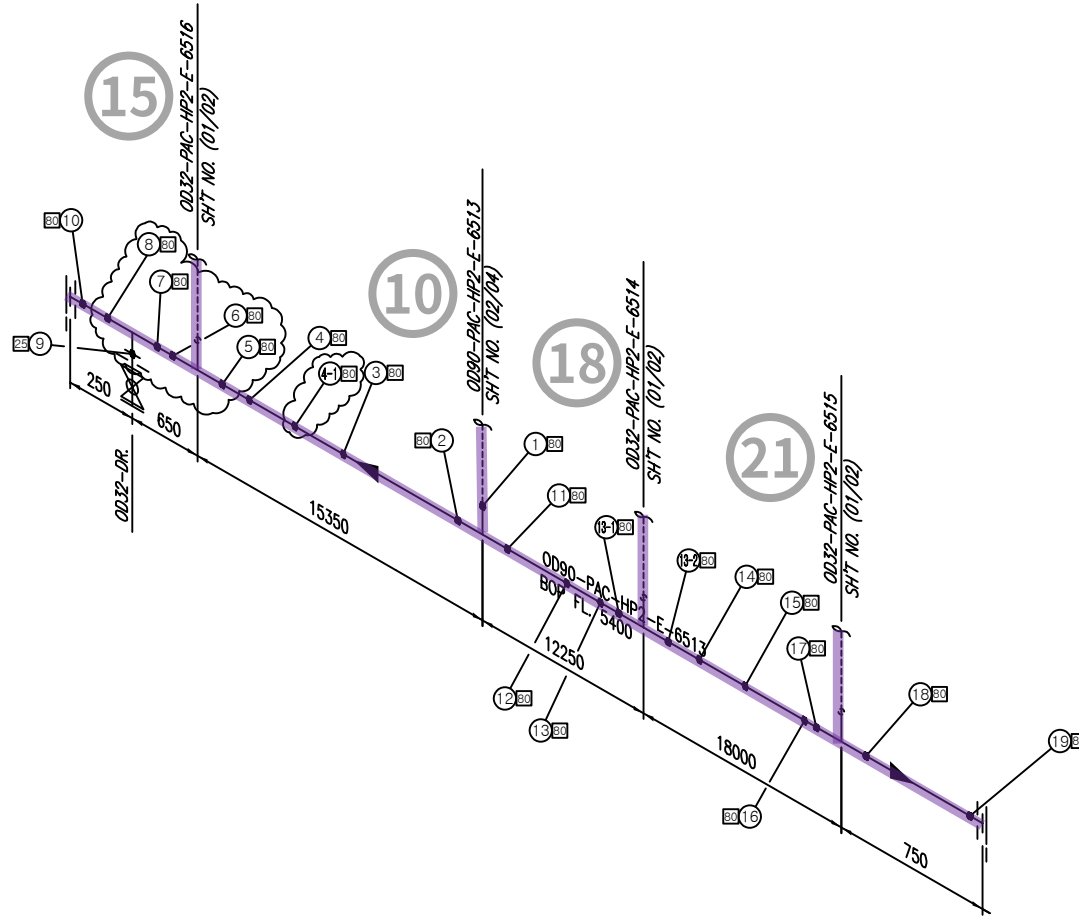
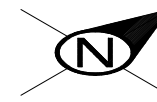
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000105		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD32-PAC-HP2-E-6558	SH'T NO. 01 / 01	REV. 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* SUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB2L * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	25	DESIGN	6.3	60	UNIT	RFA
			OPERATING	4.7	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-R10008-B	
						REF. P & ID	
						CJ-M15W14-PR-A20-000151	



25.03.31
25.01.24

REVISED AS MARKED
APPROVED FOR CONSTRUCTION

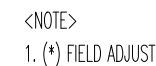
황래중 안남후 박동영
황래중 안남후 박동영

CLIENT					
CONTRACTOR				PROJECT NAME	
COMMODITY		PAC		M15X Ph4 Project	
LINE NO.		0090-PAC-HP2-E-6513		JOB NO. 22330D	
				SH'T NO. REV.	
				04 / 04 1	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



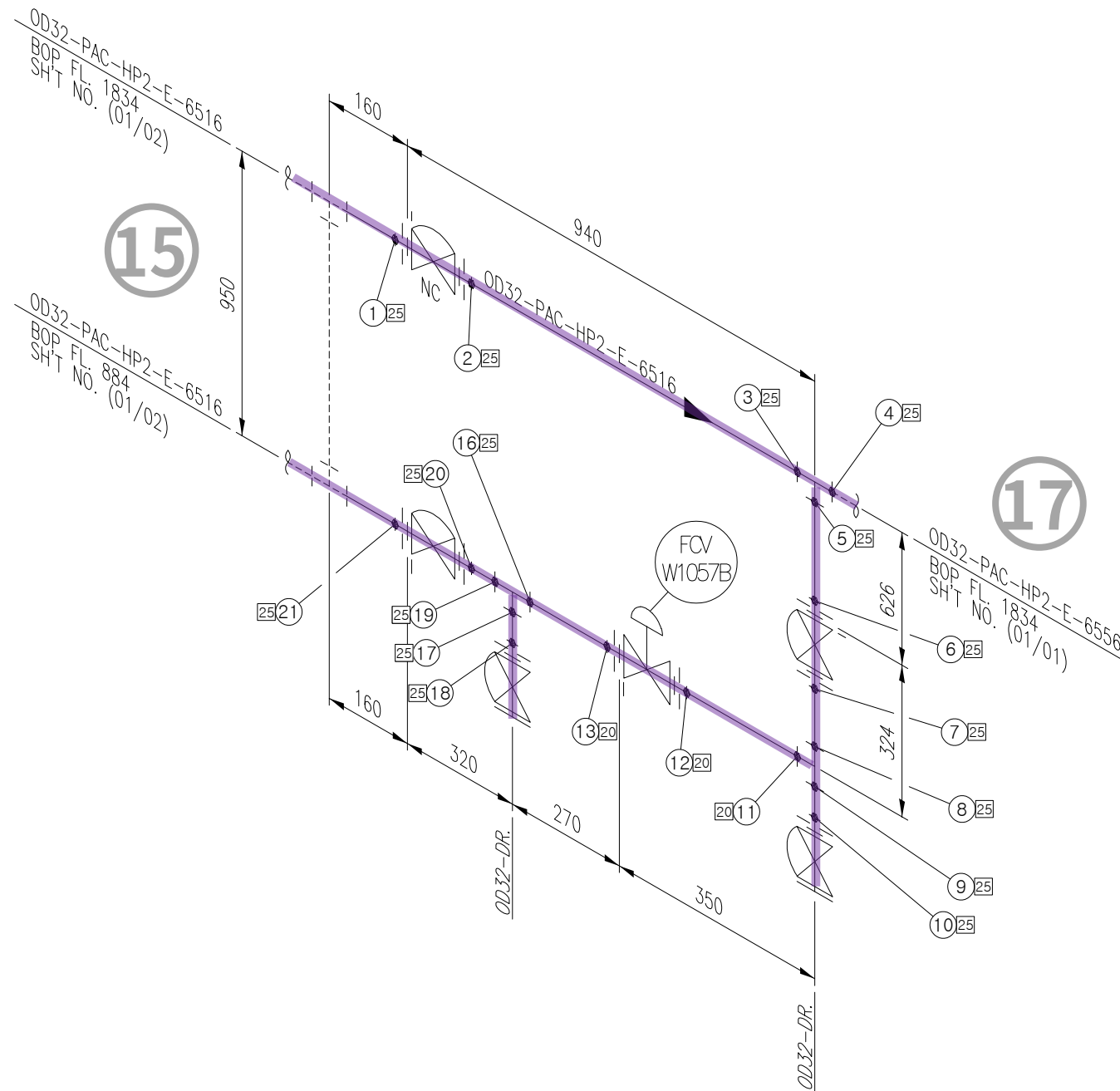
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INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000104		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 알루미늄 CASING



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL-PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT RFA							M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000104		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
													0D32-PAC-HP2-E-6516	02 / 02	1

영고 : 제 1도청은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 사용, 복제, 제3자에게 공개, 누설, 복제할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



17



18

경고 : 해당 내용은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유포를 할 수 없으며, 회사 내 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수



19



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.3	60	UNIT RFA						
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000102		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



1. (*) FIELD ADJUST

CLIENT



20

CONTRACTOR 

PROJECT NAME
M15X Ph4 Project

COMMODITY	PAC
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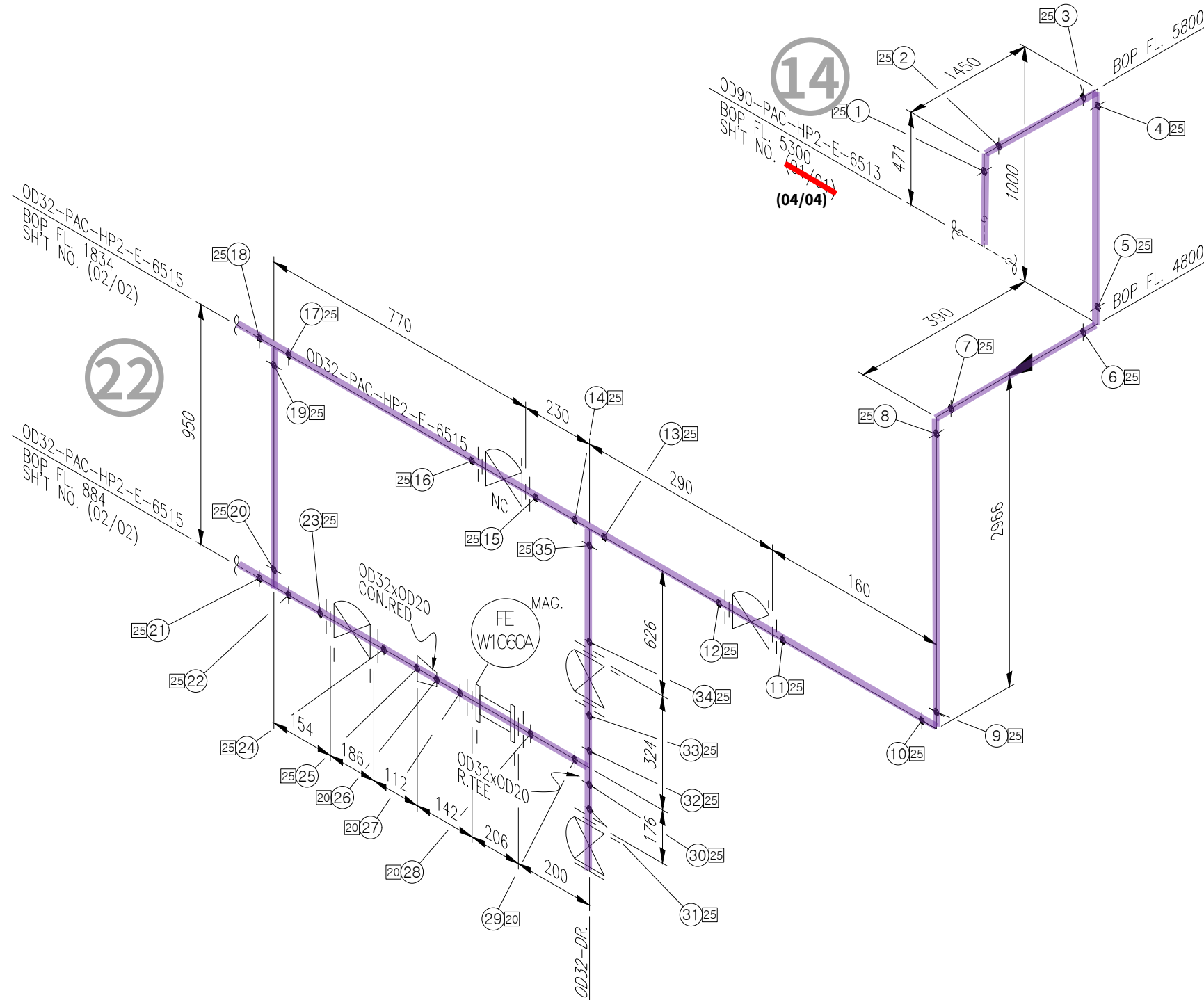
JOB NO. 22330D

LINE NO.	0D32-PAC-HP2-E-6552
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SH'T NO.	REV.
01 / 01	1



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD32	10.3M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD20	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	5EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION,
			PN16/SDR11
T	OD32	3EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD20	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD20	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION,
			PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	9EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD20	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	9EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD20	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	10SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD20	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	10.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD20	0.6M	관 보온재(고무발포) + 알루미늄 CASING



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000103		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
												0D32-PAC-HP2-E-6515	01 / 02	0

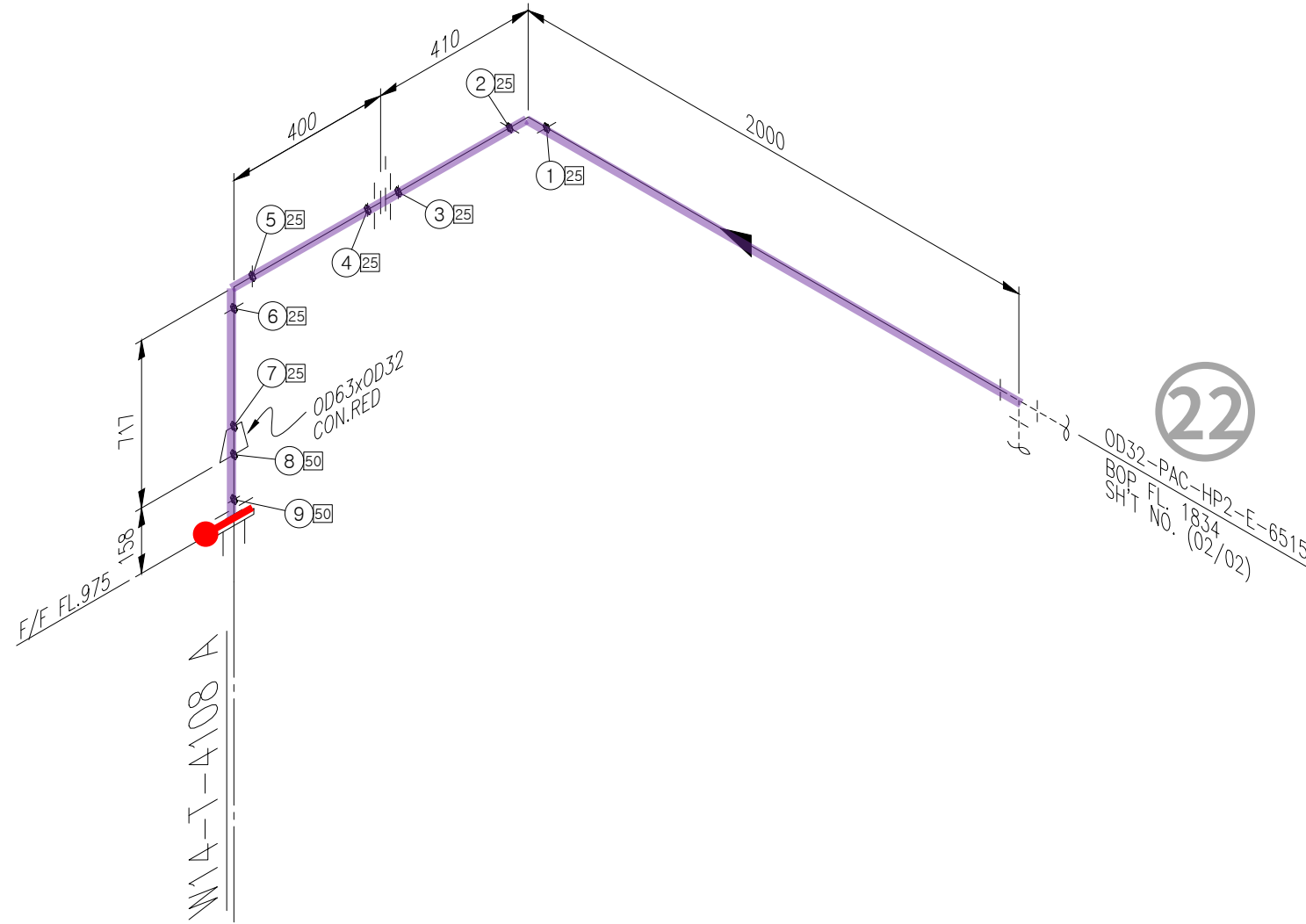
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL-PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 일루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 일루미늄 CASING

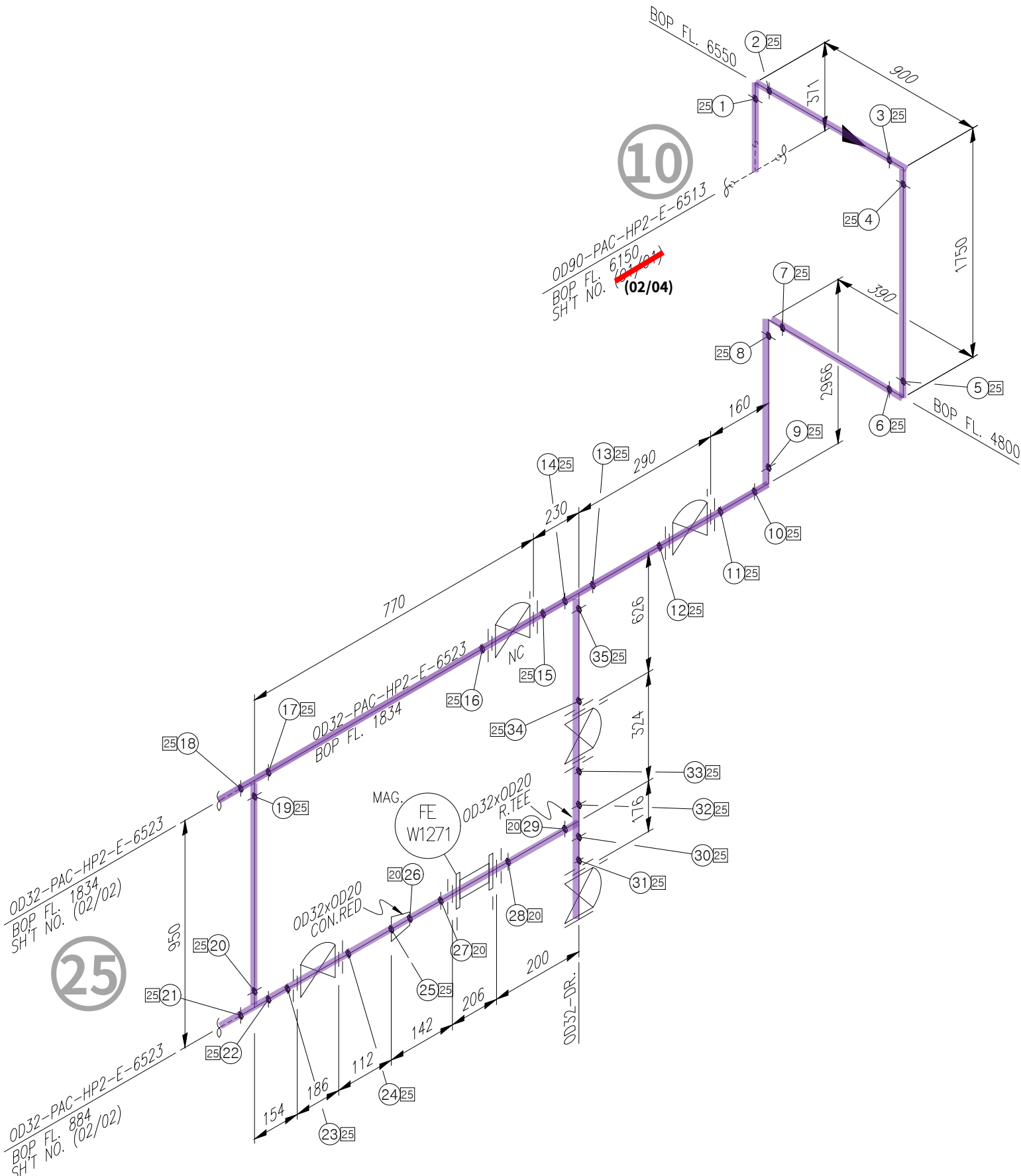
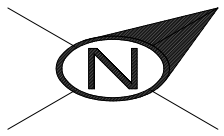
영고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 제 3자에게 공개, 제 3자에게 양도, 누설 및 무단 복제할 수 없으며, 회사나 유무자의 불법 사용으로 피해가 발생 할 경우 유무자에게 민·형사상 책임을 물을 수 있을

[illegible]

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm ² g)	TEMP. (*C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project	
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10007-A	 1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY PAC	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000103	 0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD32-PAC-HP2-E-6554	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		SH'T NO. 01 / 01	REV. 1



<NOTE>
1. (*) FIELD ADJUST

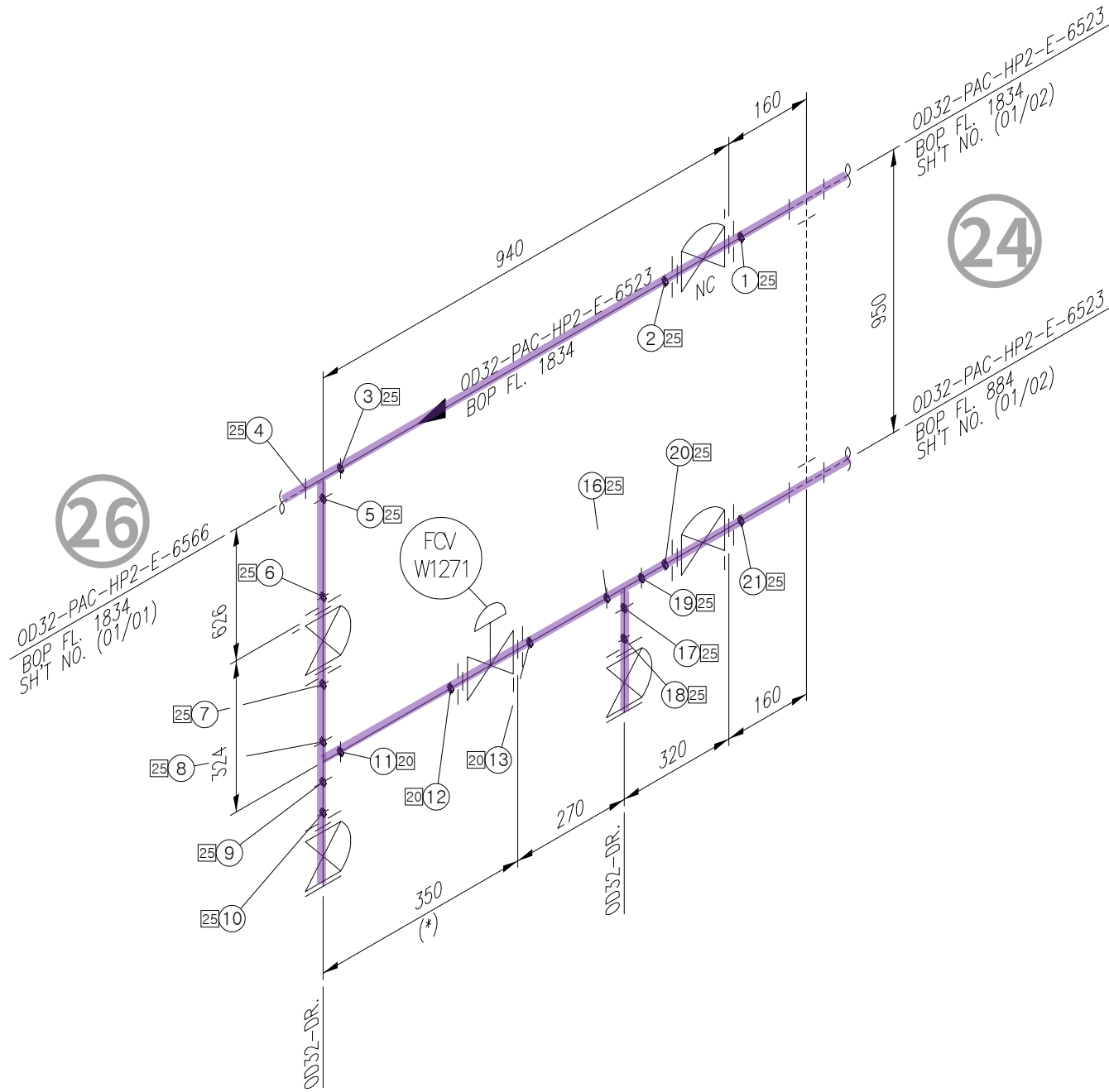
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								
	THICK.	25	DESIGN	6.3	60	UNIT	RFA								
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B		1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영		
						REF. P & ID CJ-M15W14-PR-A20-000110		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		





ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보존재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보존재(고무발포) + 알루미늄 CASING



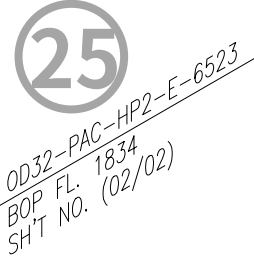
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB10/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT RFA								M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.	
								23.02.28	APPROVED FOR CONSTRUCTION	정경만	이광연	박동영			22330D	
						REF. P & ID CJ-M15W14-PR-A20-000110	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D32-PAC-HP2-E-6523	SH'T NO. 02 / 02	REV. 1

경고 : 피고인 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 독점적 외 사용, 복제, 제3자에게 공개, 누설, 정보출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있다.



1. (*) FIELD ADJUST

CLIENT



26

CONTRACTOR 

PROJECT NAME	M15X Ph4 Project
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COMMODITY	PAC
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JOB NO. 22330D

LINE NO.	0D32-PAC-HP2-E-6566
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SH'T NO. 01 / 01	REV. 1
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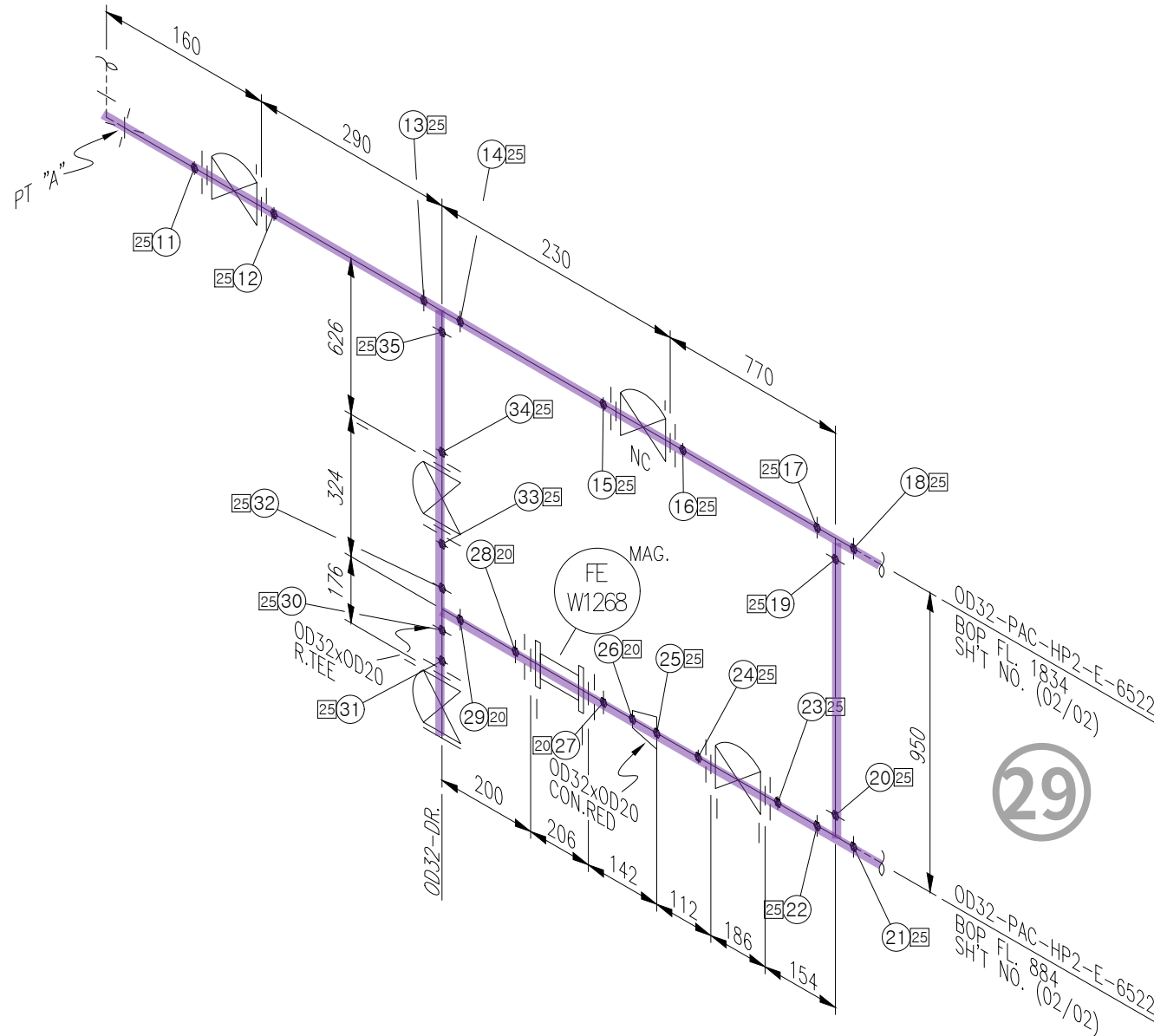


CLIENT



27

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



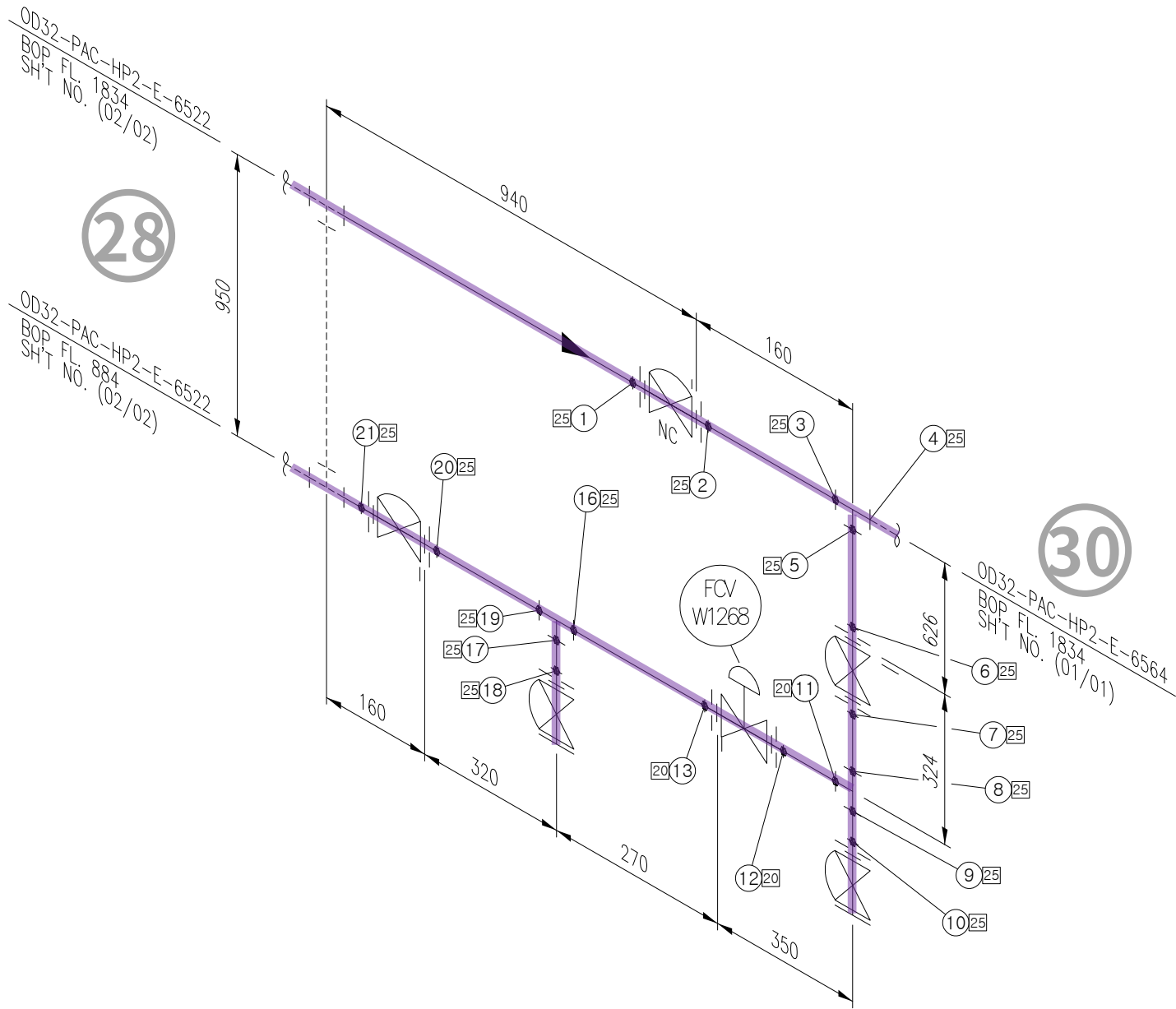
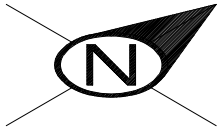
<NOTE>
1. (*) FIELD ADJUST

[illegible]

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.3	60	UNIT	RFA							M15X Ph4 Project		
			OPERATING	4.7	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	PAC	JOB NO.
						CJ-M15W14-PI-A30-R10003-B			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000109								0D32-PAC-HP2-E-6522	01 / 02	1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	6.3	60	UNIT	RFA										
			OPERATING	4.7	AMB	REF. PLAN DWG.		1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영			COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-R10003-A		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영				
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000109											
																OD32-PAC-HP2-E-6522	02 / 02
																	REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	3.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.6M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	2EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
VD	OD32	5EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD32	10EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	2EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	10EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
LOTO	OD32	1EA	LOTO
INC	25tXOD32	3.2M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tXOD25	0.6M	관 보온재(고무발포) + 알루미늄 CASING

CLIENT



CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
PAC		22330D	
LINE NO.		SH'T NO.	
OD32-PAC-HP2-E-6522		02 / 02	
		REV.	
		1	

